Least-squares and functional relations

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Abstract

In this paper, we apply the least-squares method to fit various functional models to a set of experimental data. We examine different models, including constant, linear, quadratic, and cubic forms, to determine the best fit. Two approaches are employed to solve the problem: the algebraic approach using the normal equations and the statistical approach. Both methods are used to verify the consistency of the results.

1 Introduction

The least-squares method is a mathematical technique used to approximate solutions to overdetermined systems of equations, where there are more equations than unknowns and no exact solution exists. The method minimizes the sum of squared residuals, which represent the difference between observed values and their estimates from the model.

Given a system $Ax = b$, where A is an $m \times n$ matrix, x is a vector of unknowns, and b is a vector of observations, the least-squares method provides the best approximation by minimizing the error between Ax and b.

In statistical modeling, it also helps estimate the parameters of a model that best fits a set of data by minimizing the squared differences between observed and predicted values.

2 The Least Squares Method: algebraic approach

Definition: Let $A \in R^{m \times n}$ be a given matrix, $x \in R^n$ a vector of unknowns, and $b \in R^m$ a vector of observations. The overdetermined system of linear equations is expressed as

$$Ax = b.$$

In most practical applications, an exact solution x does not exist. The least-squares solution, denoted $\hat{x}$, minimizes the error function

$$E^2 = \|Ax - b\|^2 = (Ax - b)^\top (Ax - b),$$

where $\|\cdot\|$ represents the Euclidean norm [4] and $(\cdot)^\top$ denotes the transpose [1]. Expanding E^2, we have

$$E^2 = x^\top A^\top Ax - 2b^\top Ax + b^\top b.$$

2.1 Normal Equations

Theorem. Let A be an $m \times n$ matrix and let b be a vector in R^m. The least-squares solutions of $Ax = b$ are the solutions of the matrix equation

$$A^\top Ax = A^\top b \tag{1}$$

Proof. To find $\hat{x}$, the gradient of E^2 with respect to x is set to zero

$$\nabla_x E^2 = 2A^\top Ax - 2A^\top b = 0.$$

This results in the normal equations:

$$A^\top A\hat{x} = A^\top b,$$

which are solvable when $A^\top A$ is invertible. In this case, the solution is:

$$\hat{x} = (A^\top A)^{-1} A^\top b.$$

2.2 Geometric Interpretation

The least squares solution $\hat{x}$ corresponds to the projection of b onto the column space of A. Let p represent this projection

$$p = A\hat{x}.$$

The vector p is the closest point in the column space of A to b, satisfying:

$$\|b - p\| = \min_{q \in \mathrm{Col}(A)} \|b - q\|.$$

2.3 Example

To illustrate the application of the least squares method, consider a dataset of observations.

i	1	2	3	4	5
x_i	-2	-1	0	1	2
y_i	2	4	1	-2	0

Table 1: Collected Data Points: x_i and y_i

Our goal is to identify a relationship between x_i and y_i by testing several models.

1. Constant Model ($y_i = a_0 + \epsilon_i$)
2. Linear Model ($y_i = a_0 + a_1 x_i + \epsilon_i$)
3. Quadratic Model ($y_i = a_0 + a_1 x_i + a_2 x_i^2 + \epsilon_i$)
4. Cubic Model ($y_i = a_0 + a_1 x_i + a_2 x_i^2 + a_3 x_i^3 + \epsilon_i$)

Let's represent the Matrix A for Each Model. The given data points are

$$x = [-2, -1, 0, 1, 2], \quad y = [2, 4, 1, -2, 0].$$

The first column of A is $[1, 1, 1, 1, 1]^\top$, followed by additional columns based on the powers of x_i. The matrices A for each model are:

1. Constant Model ($y_i = a_0 + \epsilon_i$)

$$A = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}.$$

with ϵ_i representing the residual error.

2. Linear Model ($y_i = a_0 + a_1 x_i + \epsilon_i$)

$$A = \begin{bmatrix} 1 & -2 \\ 1 & -1 \\ 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

3. Quadratic Model ($y_i = a_0 + a_1 x_i + a_2 x_i^2 + \epsilon_i$)

$$A = \begin{bmatrix} 1 & -2 & 4 \\ 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{bmatrix}.$$

4. Cubic Model ($y_i = a_0 + a_1 x_i + a_2 x_i^2 + a_3 x_i^3 + \epsilon_i$)

$$A = \begin{bmatrix} 1 & -2 & 4 & -8 \\ 1 & -1 & 1 & -1 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 \end{bmatrix}.$$

We apply the normal equation (1) for each model. To better understand this approach, let's consider a simple case where the model is linear

$$y_i = a_0 + a_1 x_i + \epsilon_i \tag{2}$$

For the linear model, the matrix A and vector b are:

$$A = \begin{bmatrix} 1 & -2 \\ 1 & -1 \\ 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} 2 \\ 4 \\ 1 \\ -2 \\ 0 \end{bmatrix}.$$

The least-squares solution minimizes $E^2 = \|A\hat{x} - b\|^2$. The solution is obtained by solving the normal equations (1) with

$$A^T = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ -2 & -1 & 0 & 1 & 2 \end{bmatrix}.$$

Let us first compute $A^\top A$

$$A^\top A = \begin{bmatrix} 1+1+1+1+1 & -2-1+0+1+2 \\ -2-1+0+1+2 & (-2)^2+(-1)^2+0^2+1^2+2^2 \end{bmatrix}.$$

This reduces to

$$A^\top A = \begin{bmatrix} 5 & 0 \\ 0 & 10 \end{bmatrix}.$$

Next, we compute $A^\top b$

$$A^\top b = \begin{bmatrix} 1(2)+1(4)+1(1)+1(-2)+1(0) \\ (-2)(2)+(-1)(4)+(0)(1)+(1)(-2)+(2)(0) \end{bmatrix}.$$

Hence, we find

$$A^\top b = \begin{bmatrix} 5 \\ -10 \end{bmatrix}.$$

Finally we solve $A^\top A\hat{x} = A^\top b$

$$\begin{bmatrix} 5 & 0 \\ 0 & 10 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 5 \\ -10 \end{bmatrix}.$$

Thus, we obtain

$$a_0 = \frac{5}{5} = 1, \quad a_1 = \frac{-10}{10} = -1.$$

Therefore, the least-squares solution for the linear model is

$$\hat{y} = 1 - x. \tag{3}$$

This process can be extended to the quadratic and cubic models by constructing the respective A matrices and solving the normal equations (1).

3 The Least Squares Method: statistical approach

Building on the algebraic approach, we now turn to the statistical framework, which further refines our understanding by analyzing how the data influences the parameter estimation and model fitting.

3.1 General Case

As previously stated, the least squares method is used to minimize the error between observed data and a model. Let the model be expressed as

$$y_i = a_0 + a_1 x_i + a_2 x_i^2 + \cdots + a_n x_i^n + \epsilon_i,$$

where $a_0, a_1, \ldots, a_n$ are the parameters to determine, x_i and y_i are the observed data points for $i = 1, 2, \ldots, n$, and ϵ_i represents the residual error. As mentionned previously, the objective is to find the coefficients $a_0, a_1, \ldots, a_n$ that minimize the total error E, defined as

$$E(a_0, a_1, \ldots, a_n) = \sum_{i=1}^{n} \left(y_i - (a_0 + a_1 x_i + a_2 x_i^2 + \cdots + a_n x_i^n)\right)^2. \tag{4}$$

In multivariable calculus [3], minimizing E is equivalent to solving

$$\nabla E = -2 \sum_{i=1}^{n} \left(y_i - (a_0 + a_1 x_i + a_2 x_i^2 + \cdots + a_n x_i^n)\right) \nabla \left(a_0 + a_1 x_i + a_2 x_i^2 + \cdots + a_n x_i^n\right) = 0.$$

This leads to a system of equations that can be solved to find the optimal coefficients. For simplicity, let us focus on the case where the model is linear.

3.2 Linear Regression

In the case of linear regression, the model is reduced to (1) where a_0 is the y-intercept, a_1 is the slope, and ϵ_i is the residual error. Substituting this into the error function (4), we get

$$E(a_0, a_1) = \sum_{i=1}^{n} \left(y_i - (a_0 + a_1 x_i)\right)^2.$$

To minimize $E(a_0, a_1)$, we compute the partial derivatives with respect to a_0 and a_1 and set them to zero.
First we compute the partial derivative with respect to a_0

$$\frac{\partial E}{\partial a_0} = -2 \sum_{i=1}^{n} \left(y_i - (a_0 + a_1 x_i)\right).$$

Setting this equal to zero

$$\sum_{i=1}^{n} y_i = na_0 + a_1 \sum_{i=1}^{n} x_i. \tag{5}$$

Next, we compute the partial derivative with respect to a_1

$$\frac{\partial E}{\partial a_1} = -2 \sum_{i=1}^{n} x_i \left(y_i - (a_0 + a_1 x_i) \right).$$

Setting this equal to zero

$$\sum_{i=1}^{n} x_i y_i = a_0 \sum_{i=1}^{n} x_i + a_1 \sum_{i=1}^{n} x_i^2. \tag{6}$$

From equations (5) and (6), we solve for a_0 and a_1

$$a_0 = \frac{1}{n} \sum_{i=1}^{n} y_i - a_1 \frac{1}{n} \sum_{i=1}^{n} x_i.$$

Let $\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$ and $\bar{y} = \frac{1}{n} \sum_{i=1}^{n} y_i$, we obtain

$$a_0 = \bar{y} - a_1 \bar{x}. \tag{7}$$

Substituting a_0 into equation (6) we obtain

$$a_1 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}. \tag{8}$$

The regression line is thus

$$y = a_0 + a_1 x. \tag{9}$$

3.3 Example: Solving with Data

Given the data points from the previous example.

$$x = [-2, -1, 0, 1, 2], \quad y = [2, 4, 1, -2, 0],$$

we compute the required values step by step.

Let us first compute the means

$$\bar{x} = \frac{-2 + (-1) + 0 + 1 + 2}{5} = 0, \quad \bar{y} = \frac{2 + 4 + 1 + (-2) + 0}{5} = 1. \tag{10}$$

Next, we calculate a_1

$$a_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} = \frac{(-2)(1) + (-1)(3) + (0)(0) + (1)(-3) + (2)(-1)}{(-2)^2 + (-1)^2 + 0^2 + 1^2 + 2^2}.$$

This reduces to

$$a_1 = \frac{-2 - 3 + 0 - 3 - 2}{4 + 1 + 0 + 1 + 4} = \frac{-10}{10} = -1.$$

Thus we obtain a_0:

$$a_0 = \bar{y} - a_1\bar{x} = 1 - (-1)(0) = 1.$$

$$a_1 = -1, \quad a_0 = 1.$$

Therefore the regression line is

$$y = -x + 1.$$

This confirms the results found in the first example using the first method. float

4 Python program

We use Python to solve the least-squares optimization for the linear model. This confirms that the regression equation is indeed

$$y = -x + 1,$$

thereby confirming our previous results using both methods. We visualize this numerically in Figure 1.

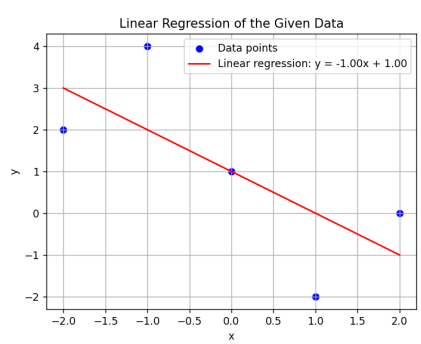

Figure 1: Visualization of the linear regression model $y = -x + 1$

5 Conclusion

In this paper, the least-squares method has been shown to provide a powerful tool for fitting functional models to experimental data, allowing us to estimate parameters and minimize the error between observed and predicted values. By applying both algebraic and statistical approaches, we ensure the robustness and consistency of the results. Ultimately, least squares offers a reliable framework for data analysis in various fields, from simple linear regression to more complex polynomial models.

References

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Appendix

Below is the Python code used to perform the least-squares linear regression and visualize the data along with the regression line:

```python
import numpy as np
import matplotlib.pyplot as plt

# Given data from the table
x = np.array([-2, -1, 0, 1, 2])  # x-values
y = np.array([2, 4, 1, -2, 0])   # y-values

# Perform linear regression using np.polyfit
# polyfit returns the coefficients of the polynomial fit (degree 1 for linear)
coefficients = np.polyfit(x, y, 1)  # Degree 1 for linear regression
slope, intercept = coefficients

```

```python
# Generate the fitted y-values using the equation y = slope * x + intercept
y_fit = slope * x + intercept

# Plotting the data and the regression line
plt.scatter(x, y, color='blue', label='Data points')  # Scatter plot for data
plt.plot(x, y_fit, color='red', label=f'Linear regression: y = {slope:.2f}x + {intercept:.2f}')
plt.xlabel('x')
plt.ylabel('y')
plt.title('Linear Regression of the Given Data')
plt.legend()
plt.grid(True)
plt.show()

# Print out the regression line equation
print(f"Regression line: y = {slope:.2f}x + {intercept:.2f}")
```